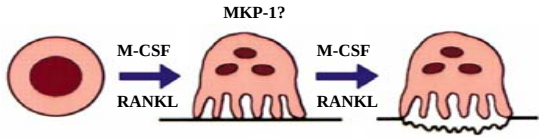
Understanding the Role of Bone Morphogenic Proteins in MAPK Phosephatase-1 (MKP-1) Regulation During Osteoclastogenesis

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Introduction

- Bones are dynamic tissues that are constantly undergoing remodeling. Two cell types important in bone remodeling are osteoblasts that form bone and osteoclasts, which resorb bones and are the focus of this study[1].
- Two proteins that regulate osteoclast differentiation and are subjects of interest in the study are BMP and MKP-1.
- Bone morphogenic proteins (BMPs) are growth factors that have been shown to enhance osteoclast differentiation[2].
- MAPK phosphatase-1 (MKP-1) is a protein that regulates MAP Kinase activation. MAP Kinases are necessary for osteoclast differentiation[3].



Methods

Osteoclast Differentiation Real-Time RT-PCR

- Bone marrow cells were harvested from 6-week-old wild type mice.
- The cells were treated with Osteoclast and CMG (1%) conditioning media containing (M-CSF) for 3 days.
- The cells were treated with either RANKL alone (20ng/mL) or RANKL+BMP2 (30ng/mL) in triplicate for each time-point.
- RNA was isolated, cDNA synthesized and gene expression was analyzed by Real-Time RT-PCR.

Transfection

3T3

- The NIH 3T3 cells were transfected 300ng of MKP-1-luc reporter or BRE-luc reporter.
- The cells were transfected with 250ng, 500ng and 1µg concentrations of constitutively active SMAD1 protein.

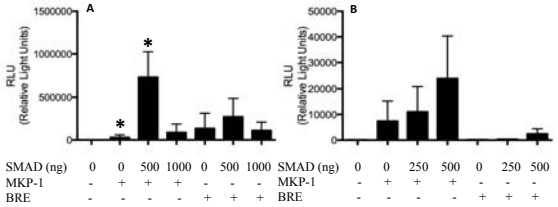
C2C12

- The C2C12 cells were transfected with BRE-luc or MKP-1-luc reporters at a concentration of 300ng.
- The cells were then treated in media containing various concentrations of BMP2 ranging from 0 to 100ng/ml.

Results

SMAD1 Activates MKP-1 Promoter

- SMAD1 significantly increased MKP-1 promoter activation; the addition of 500ng of SMAD1 increased luciferase expression 20-fold over reporter alone.
- The dose-dependent response was observed in Figure A. and then reproduced at lower concentrations of SMAD1.



SMAD (ng)	0	0	500	1000	0	500	1000
MKP-1	-	+	+	+	-	-	-
BRE	-	-	-	-	+	+	+

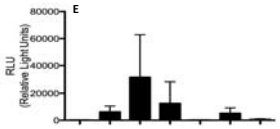
SMAD (ng)	0	0	250	500	0	250	500
MKP-1	-	+	+	+	-	-	-
BRE	-	-	-	-	+	+	+

Figure1. A) 3T3 cells were transfected with SMAD1 and either MKP-1-luc and BRE-luc reporter plasmids. The luciferase expression was measured. *p< 0.05.

B) 3T3 cells were transfected with reduced concentration of SMAD1 and either MKP-1-luc and BRE-luc reporter plasmids. The luciferase expression was measured.

BMP2 Activates MKP-1 Promoter

- MKP-1 promoter activity increased in C2C12 cells when treated with BMP2.
- At 50 ng concentration of BMP2 exhibited the highest level of MKP-1 activation.



BMP2 (ng)	0	0	50	100	0	50	100
MKP-1	-	+	+	+	-	-	-
BRE	-	-	-	-	+	+	+

Figure 3. E) C2C12 cells were transfected with either MKP-1-luc and BRE-luc reporter plasmids. The cells were treated with BMP2 and the luciferase expression was measured.

MKP-1 Shows a Delayed Expression During Differentiation

- Cathepsin K, a positive control, RT-PCR data showed an increased of RNA expression on days 2 and 3 in both RANKL and RANKL with BMP2 treated cells.
- MKP-1 expression detected in Real-Time RT-PCR showed:
  - An increase in cells treated with RANKL in day 2 and a significant decrease the following day.
  - Cells treated with both RANKL and BMP2 followed an opposite trend showing RNA activation on day 3.

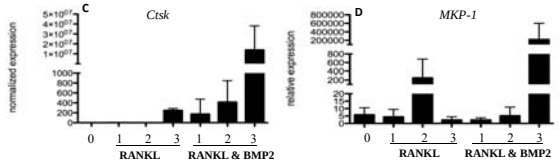


Figure 2. C) The osteoclast wild type cells were treated with media containing RANKL alone or RANKL plus BMP2 at each time-point. Ctsk primer was used to measure RNA expression.

D) The osteoclast wild type cells were treated with media containing RANKL alone or RANKL plus BMP2 at each time-point. mMKP-1 primer was used to measure RNA expression.

Conclusion

- This study has shown that BMP plays a role in MKP-1 expression in osteoclastogenesis.
- SMAD1, a factor activated by bone morphogenic proteins, was shown to activate the MKP-1 promoter in a dose responsive manner.
- PCR findings demonstrate that upregulation of MKP-1 expression is delayed in osteoclasts treated with RANKL and BMP2 compared to RANKL alone.
- The results suggest that BMP signaling modulates expression of MKP-1 both positively or negatively depending on context – during osteoclast differentiation, BMPs delay expression of MKP-1, whereas in a reporter assay, activated SMAD1 stimulates MKP-1 promoter activity.

References

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2) Jensen, E.D., Pham, L., Billington, C.J., Espe, K., Carlson, A.E., Westendorf, J.J., Petryk, A., Gopalakrishnan, R., Mansky, K., 2010. Bone morphogenic protein 2 directly enhances differentiation of murine osteoclast precursors. Journal of cellular biochemistry 109, 672–682.
3) Carlson, J., Cui, W., Zhang, Q., Xu, X., Mercan, F., Vignery, A., 2009. Role of MKP-1 in Osteoclasts and Bone Homeostasis. The American Journal of Pathology 175, 1564–1573.

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http://genesdev.cshlp.org/content/17/18/2205/F8.expansion.html&usg=ALkJrhiCdWT1O6xRr22gauXdbEqvgafqTA

Concept Map

